

Representation Convergence — Results Summary

Every experiment's measured outcome in one place. Thesis: independently trained models share representations up to a linear transformation. All figures are held-out; controls and chance rates are quoted alongside. Full method and caveats in [Final_Project_Report.pdf](#) (52 pp).

The findings at a glance

#	finding	headline evidence
1	Convergence is linear	MLP = ridge to 4 decimals (0.9183 = 0.9183), reproduced on two independent runs of a fresh corpus
2	Related, not rigid	Procrustes R2 0.038 vs ridge 0.592; 69x spread in the map's singular values
3	Deployable	95% of ceiling at 0.79 MB; verification AUC 0.999
4	Cross-modal without joint training	DINOv2 -> bge 40.9-58.3% of a matched ceiling, up to 358x chance
5	Alignment scales with capacity	+15.3 points of ceiling per decade of parameters, six monotone measures
6	Isotropy, not similarity-training	whitening a raw LM space reaches 102.6% / 101.4% of a contrastive encoder at two scales
7	Correspondence is self-identifying	Gromov-Wasserstein 39.7% exact = 595x chance, zero paired examples
8	One shared space, used	hub transfer 92.9-96.5% of native on encoders never seen, incl. cross-lineage
9	The mechanism predicts	pre-registered encoder ladder, rank correlation +1.00

Meta-finding, measured three times. Isotropy helps when READING a correspondence (C.11), hurts when FINDING one (C.12), and hurts when CARRYING several at once (the hub width cliff). One geometric property, three opposite prescriptions.

Experiment A — scientific proof (GPT-2 vs Pythia-160M)

metric	fail	pass	measured
CKA diagonal mean (trained)	<0.15	>0.35	0.424
CKA mean, random control	>0.20	<0.15	0.067
trained / random ratio	<2x	>=3x	6.3x
stitching R2, peak layer (L11)	<0.50	>0.70	0.797
stitching R2, random control	>0.40	<0.30	0.173
depth-slope signature	both flat	opposite	0.59->0.80 vs 0.38->0.09

Verdict: PASS. The trained curve rises with depth while the random control decays - opposite slopes, an 8x gap at L11. Run 1 (1,646 samples) failed on sample starvation; run 2 (10,000) passed decisively, identical pipeline.

Experiment B — the deployed adapter (MobileCLIP-S1 -> SigLIP 2)

quantity	value
Ridge adapter: held-out R2 / mean cosine	0.592 / 0.900
Procrustes (rotation only): R2	0.038 (anisotropy, not just orientation)
MLP adapter: best cosine	0.903 (+0.003 -> relationship is linear)
Retrieval R@1 / R@5 / R@10 (ceiling)	0.630 / 0.885 / 0.944
Retrieval with linear adapter	0.557 / 0.843 / 0.918
as % of ceiling	88.4% / 95.3% / 97.2%
raw cross-space baseline	0.001 / 0.005 / 0.010 (exact chance)
verification / ranking / membership	AUC 0.999 / R@1 0.987 / AUC 0.889
hubness N10 skew: raw -> adapted	4.28 -> 1.61 (ceiling 1.41)
deployed artifact	Adapter.mlpackage, 0.79 MB fp16, iOS 16+

Series E — cross-modal, scaling, geometry

experiment	result
E1 capacity sweep (DINOv2 -> bge)	small/base/large R@1 40.9% / 53.5% / 58.3% of ceiling; +15.3 pts per decade, fit R2 0.949
E1.1 swap text encoder to GPT-2	R2 retains 71%, retrieval collapses to 34%; cosine 1.000 while R@1 0.160
E1.2 isotropy rescue	whitening lifts R@1 0.130 -> 0.478 = 102.6% of a 1.2B-pair contrastive encoder; eff. rank 6.5 -> 490.8; iso-retrieval corr +0.94
E1.3 encoder ladder (pre-registered)	P1 rank correlation +1.00 (confirmed); P2 partial - isotropy necessary, not sufficient

Series F — the contrast condition (deliberate training)

experiment	result
F1/F2 capacity test	MLP converges to EXACTLY linear: cosine 0.9183 = 0.9183, gain +0.0000, reproduced on two runs from a fresh 38k-row corpus (74.2 rows/dim)
F5 data test	linear refit on 38k gives R@1 +0.0244 (bootstrap CI excludes 0); 90%-of-ceiling gate reported NOT met at 89.0%
F3 unfreeze bottleneck	built and gated shut - running it would cost the clean 'no pretrained weight modified' claim; the gate is the contribution

Series C — pair-free recovery

quantity	value
Gromov-Wasserstein exact matches	39.7% = 595x chance, zero pairs
map fitted from that assignment	R@1 0.252 (40% of ceiling), verification AUC 0.906
negative 1: ICP refinement	degrades monotonically (converges on its own errors)
negative 2: whitening before matching	39.7% -> 0.1% (destroys matching)
scope (C.12.1)	both encoders contrastive image-text - the WEAKEST-independence pair; result is bounded accordingly

Series G — the shared hub

experiment	result
G1 seven-space hub, transfer	R@1 0.451 at HUB_DIM 512 (AUC 0.986); transfer cliff at 512->768; vs 0.445 at five spaces - UNCHANGED (within noise), which is the useful result
G1 composition (scale-free R@1)	95.6% retention over 36 of 41 pairs; the 5 exceptions all write INTO GPT-2
G1 read/write asymmetry	GPT-2 as source 95.1%, as target 24.8% - a collapsed space is readable out of, not writable into; tracks effective rank across bge/BERT/GPT-2
G3 compatibility (vs bespoke hubs)	42 of 42 ordered pairs SHARED, mean retention 103.6%; control at chance (0.004 vs 0.001)
G4 cross-lineage transfer	SigLIP 2 (independent lineage) 94.2% of native - INSIDE the 93.8-96.5% within-family band; control at chance
G5 scale of agreement	ratio-to-chance falls for all 21 pairs (967x at k=1 -> 2.3x at k=500): agreement is LOCAL, global structure diverges
G5 what explains it (exploratory)	objective (+53x) > lineage (+44x) > modality (+29x); top pair bge-SBERT shares objective, no lineage

Series H / V — certification

check	result
H1 artifact battery	47 checks, NO hard failures; 2 flags, both the documented GPT-2 collapse, reproduced independently in two files
rows per input dimension	every fit 5.9-9.8, above the pre-registered floor of 5
alignment (shuffle test)	honest fits beat shuffled by 0.545-0.863 vs a 0.2 threshold

V1 independent reproduction	verification AUC 0.9994, R@1 0.987, 69x knob spread - three reported figures from a separate code path
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Nine findings, five documented self-corrections reached by measurement rather than argument, and a theory that survived its own out-of-sample test. Every number here is held-out and quoted against a control.